

SpectralBridge: Making Cross-Sensor Reflectance Translation Inspectable

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Abstract

Observing the same ground does not make reflectance measurements from different instruments equivalent. Drone, airborne, and satellite observations integrate different wavelengths, spatial areas, and acquisition geometries. SpectralBridge makes the transformations among these measurement spaces explicit. Its canonical NEON workflow exports hyperspectral observations, applies topographic and bidirectional reflectance distribution function (BRDF) normalizations, integrates spectra through configured sensor responses, produces tabular products, and records stage-level quality-assurance evidence. Named intermediate files preserve the states between source observation and translated value, allowing disagreement to be traced to a specific operation rather than left as an unexplained residual. A documented NEON flightline completed end to end while retaining an overall warning, demonstrating that execution and confidence are reported separately. Across 100 flightlines, *[pending]* completed all declared stages; combined reports were PASS for *[pending]*, WARN for *[pending]*, and FAIL for *[pending]*. SpectralBridge demonstrates an inspectable, restartable workflow for constructing cross-sensor comparisons while leaving empirical sensor agreement to be tested against independent observations.

1 Introduction: the same ground is not the same measurement

Imagine a vegetation patch observed by a MicaSense-equipped drone, the NEON Airborne Observation Platform, and Landsat. The drone may isolate a sunlit crown from nearby soil and shadow. NEON records a nearly continuous spectrum at meter scale. Landsat integrates a much larger area into several broad bands. Even when all three products contain a value called “red” or “near infrared,” those values are not interchangeable.

The instrument helps define the measurement. Its spectral response weights particular wavelengths; its pixel footprint mixes particular objects; and its viewing and illumination conditions shape directional reflectance [2, 4, 1]. The same place is therefore not necessarily the same measurement. *Comparable reflectance must be constructed, not assumed.*

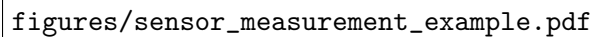
This distinction matters whenever fine-scale ecological observations are linked to satellite records. A crown-level relationship can weaken when soil and shadow enter a coarse footprint; a red-edge

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feature can shift when a broad response averages across it; and an illumination pattern can resemble a biological gradient. Without an explicit translation, those effects arrive together in the final comparison.

SpectralBridge addresses this problem by treating comparison as a sequence of visible translations. That visibility changes what a scientist can learn from disagreement. A black-box harmonization produces a residual. An inspectable translation lets the researcher ask whether that residual entered with the source observation, geometry normalization, spectral weighting, spatial aggregation, tabulation, or a real ecological difference.

Figure 1 anchors both parts of the problem. A coarse footprint mixes materials that a drone can separate, while different red-band responses produce different values from the same illustrative spectrum. The example is conceptual, but the band centers and widths come from SpectralBridge’s packaged definitions.



figures/sensor_measurement_example.pdf

Figure 1: How the same landscape produces different measurements. The upper panel shows how increasingly large footprints mix vegetation, soil, and shadow. The lower panel applies illustrative response profiles, parameterized by the packaged Landsat 8 OLI and MicaSense red-band centers and full widths at half maximum, to one illustrative spectrum [15]. The resulting values demonstrate spectral weighting, not empirical sensor agreement. Generated by `scripts/make_sensor_figure.py`.

2 A translation problem, not a conversion

A cross-sensor workflow must answer five practical questions: What did each sensor measure? What must change before comparison? Which operations changed the observation? Can those operations

be inspected and repeated? When did the software finish while the result still warrants caution?

These questions separate three tasks. Geometry normalization addresses variation associated with terrain, illumination, and viewing direction. Spectral translation asks how a dense spectrum averages within a target response. Spatial reconciliation determines how fine-scale objects contribute to a coarser footprint. SpectralBridge implements the first two and preserves geospatial products for the third [9].

Calling this process a translation, rather than a conversion, keeps those tasks distinct. A spectral response can be applied without pretending that the footprint changed. Geometry can be normalized without erasing the acquisition conditions from the record. Spatial aggregation can then be designed for the ecological question instead of being hidden inside a generic resampling step.

NEON is the translating reference in the project-level relationship Drone $\rightarrow$ NEON $\rightarrow$ Landsat. Its dense spectra and ancillary geometry support transformations unavailable from a few broad bands alone. The drone path remains separate, preserving native names, provenance, and correction status. Corrected and sensor-translated NEON products then provide a reference for analyses that reconcile acquisition time, co-registration, and footprint aggregation [8, 10].

The evidence boundary is explicit. *Execution* means that a stage produced its declared artifacts. *QA* means that an evaluated diagnostic passed, warned, or failed a stated rule. *Scientific validation* requires comparison with independent observations or a justified external reference. SpectralBridge’s contribution is the inspectable organization of established correction and resampling methods, not a claim of universal sensor calibration.

3 What changes before comparison

Spectral response

Different instruments weight different parts of the spectrum. For reflectance $\rho(\lambda)$ and target response $s_b(\lambda)$, SpectralBridge computes the corresponding response-weighted band value,

$$\hat{\rho}_b = \frac{\int \rho(\lambda) s_b(\lambda) d\lambda}{\int s_b(\lambda) d\lambda}. \quad (1)$$

using response arrays sampled on the source wavelength grid. Controlled tests verify the discrete calculation to an absolute tolerance of 2×10^{-7} [13, 20]. The result answers a precise question: what band value would the NEON spectrum produce under the configured response? It does not alter the source footprint or reproduce the atmosphere, timing, and processing of an actual satellite acquisition.

Geometry and terrain

The same surface can appear different as the sun, sensor, or terrain changes. SpectralBridge applies sun–canopy–sensor plus C (SCS+C) topographic normalization followed by kernel-based BRDF normalization. By default, one SCS+C term per band is estimated across the flightline before scene-level BRDF coefficients are applied with local geometry [3, 14]. An optional tiled fit

makes grid-aligned seams a specific pattern to inspect. These operations reduce modeled geometry dependence; their scientific benefit for a particular comparison remains an empirical question.

Spatial support

Spectral convolution leaves spatial support unchanged. A crown-scale spectrum and a Landsat footprint still represent different mixtures after they occupy similar band spaces. SpectralBridge preserves geospatial raster and polygon-extraction products so co-registration and aggregation can be performed explicitly; the choice of aggregation model remains part of the scientific analysis.

For example, a hypothetical satellite footprint containing vegetation, exposed soil, and shadow should be compared with the corresponding fine-scale mixture, not with a conveniently selected crown. In that case, the aggregation rule is part of the measurement definition: it determines which fine-resolution observations contribute, how missing areas are handled, and whether the comparison represents area, cover, or another ecological quantity.

4 An inspectable pipeline

The canonical NEON path is shown in Figure 2. Each stage writes a named artifact that captures a distinct measurement state.

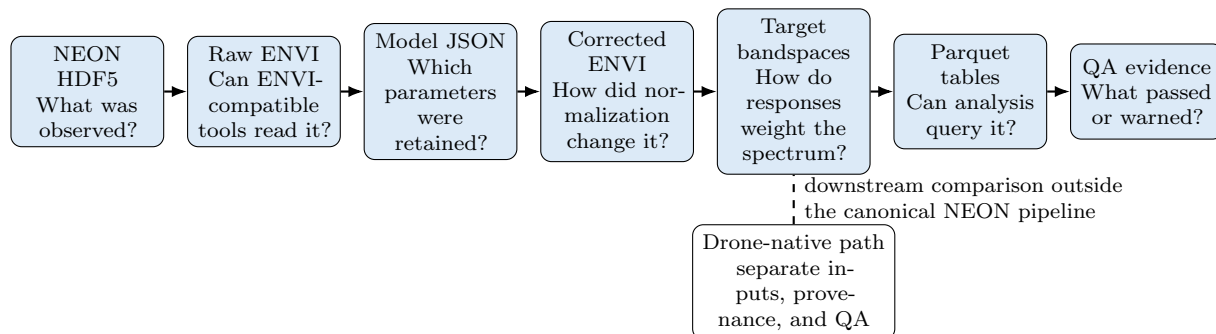


Figure 2: The implemented NEON workflow is an ordered file pipeline. The dashed connection marks downstream comparison with drone products, not an implemented direct conversion. The canonical corrected cube combines topographic and BRDF normalization; separate corrected cubes are not retained.

The source HDF5 is retained, and raw reflectance is exported to an ENVI image/header pair. A JSON sidecar records the BRDF model and geometry summaries before the pipeline writes the combined topographic-and-BRDF-normalized cube. The normalized cube is then convolved into configured Landsat TM, ETM+, OLI, OLI-2, and MicaSense products. Each product answers how the corrected NEON spectrum averages under that band definition.

Each ENVI product receives a Parquet sidecar, and DuckDB assembles a per-flightline merged table. QA writes visual panels and machine-readable metrics and reports. Canonical paths and per-flightline directories keep these products associated during parallel and restarted runs [11, 12, 17, 7].

The sequence matters scientifically. The raw ENVI product is the baseline observation; the corrected

cube records the result of geometry-related modeling; the target-sensor products record spectral weighting; and the Parquet tables expose those states to statistical analysis without replacing the spatial rasters. A researcher can therefore move between mapped patterns, spectra, fitted models, and analysis rows without losing the link among them.

5 Outputs are the API

SpectralBridge describes its file-mediated architecture as “outputs are the API.” Intermediate products are not disposable cache; they preserve the observation as it moves through different measurement spaces. A black box exposes input and final value. SpectralBridge exposes raw observation $\rightarrow$ corrected observation $\rightarrow$ target-sensor representation $\rightarrow$ extracted table, with QA evidence beside each stage [18].

Suppose a translated red band contains an unexpected spatial pattern. Was it already present in the raw cube? Did it appear after geometry normalization? Was it amplified by the target response? Did extraction change the sampled records? The preserved states turn one residual into a set of testable hypotheses about a named transformation. This is the central scientific payoff of the architecture: SpectralBridge helps investigate *why* sensors disagree, not only observe that they disagree.

The same files make long hyperspectral runs restartable. A rerun reuses artifacts that pass completeness and readability checks and rebuilds missing or malformed outputs [19, 21]. This mattered across the flightline cohort: source HDF5 files ranged from approximately 2.5–16 GB, and the more than sixfold spread produced substantial differences in RAM demand. A single memory recommendation would therefore be misleading; resource planning should relate peak resident memory to input size. Reproducible reuse still requires the code version, configuration, input identifiers or checksums, and packaged response definitions to travel with the outputs. The canonical corrected cube also combines topographic and BRDF effects, so separate attribution requires an additional analysis.

6 Quality assurance across flightlines

R10C provides the simplest QA story: the workflow completed, but the combined interpretation remained WARN. Every declared stage produced its artifacts. Acquisition, spectral convolution, and table readability passed their configured checks; input reflectance, correction parameters, and combined correction warned [23, 24].

Why did a successful run warn? Input QA retained 68 declared poor-quality bands rather than silently replacing them. Four persisted geometry summaries fell outside expected radian ranges and were flagged for review. The report also kept scientific comparisons that had not been performed—including direct Landsat comparison and separate attribution of topographic and BRDF effects—in a visible “not evaluated” state. R10C therefore demonstrates the intended distinction: completion describes execution; PASS, WARN, and FAIL describe evaluated rules; scientific confidence depends on the evidence available for the intended comparison.

The same logic structures the 100-flightline analysis. Each flightline is the unit of evaluation. We summarize artifact completion; combined and stage-level status counts; the prevalence of geometry-

range warnings; distributions of the input fraction above 1.2, correction magnitude, and convolved fraction above 1.2; and readable Parquet outputs. Because source files ranged from approximately 2.5–16 GB, we also report peak resident memory by flightline and examine its relationship with input size rather than pooling all runs into one RAM estimate. Status denominators include all flightlines, with missing or unreadable reports counted explicitly rather than dropped. Continuous diagnostics are summarized across flightlines rather than pooled across pixels, preventing large cubes from dominating the result. Site and acquisition year remain available for stratified checks when the cohort composition warrants them.

Table 1: Planned summary of QA evidence across 100 NEON flightlines. Counts should be reported with percentages; continuous diagnostics should be reported as median [interquartile range] and, where useful, range.

Measure	Cohort result
All declared stages produced readable artifacts	[pending]
Combined status: PASS	[pending]
Combined status: WARN	[pending]
Combined status: FAIL	[pending]
Input-reflectance warnings	[pending]
Correction-parameter warnings	[pending]
Combined-correction warnings	[pending]
Spectral-convolution warnings or failures	[pending]
Analysis-table warnings or failures	[pending]
Flightlines with geometry-range warnings	[pending]
Input fraction above 1.2	[pending]
Correction absolute-difference 99th percentile	[pending]
Convolved fraction above 1.2	[pending]
Readable Parquet outputs per flightline	[pending]
Source HDF5 file size, range	2.5–16 GB
Peak resident memory by flightline	[pending]
Relationship between input size and peak memory	[pending]

Across 100 flightlines, [pending] completed all declared stages. Combined QA reports were PASS for [pending], WARN for [pending], and FAIL for [pending]. The most common warning stage was [pending]. Source files spanned 2.5–16 GB; peak resident memory was [pending], with an input-size relationship of [pending]. These cohort results show whether R10C’s warning pattern is typical or exceptional while also defining realistic resource requirements for flightlines of different sizes.

7 Scope of the evidence

Current evidence supports the software claims central to this note: SpectralBridge executes explicit geometry and spectral transformations, preserves intermediate states, creates raster and tabular products, resumes validated stages, and reports warnings and unevaluated diagnostics. Deterministic tests exercise arithmetic, file-writing, restart, and reporting contracts [22]. The flightline cohort characterizes operational behavior across runs; paired sensor observations and independent references are still required to evaluate calibration accuracy, ecological benefit, and transfer across ecosystems [25].

8 Discussion

SpectralBridge makes cross-sensor disagreement diagnosable. Because the raw, normalized, spectrally translated, and tabulated states remain available, a researcher can locate where an unexpected pattern first appeared and formulate hypotheses about geometry, spectral response, spatial support, processing, or the landscape itself. The result is more than a harmonized value: it is an inspectable account of how that value was constructed.

The 100-flightline analysis establishes how reliably this account is produced and which warnings recur. The next validation step is different: paired drone–NEON observations, direct Landsat comparisons, explicit spatial aggregation, and uncertainty analysis are needed to test whether particular translations improve agreement or inference. Those evaluations extend the evidence without changing the design principle: comparable reflectance is an examined result, not a consequence of observing the same ground.

9 Availability and development status

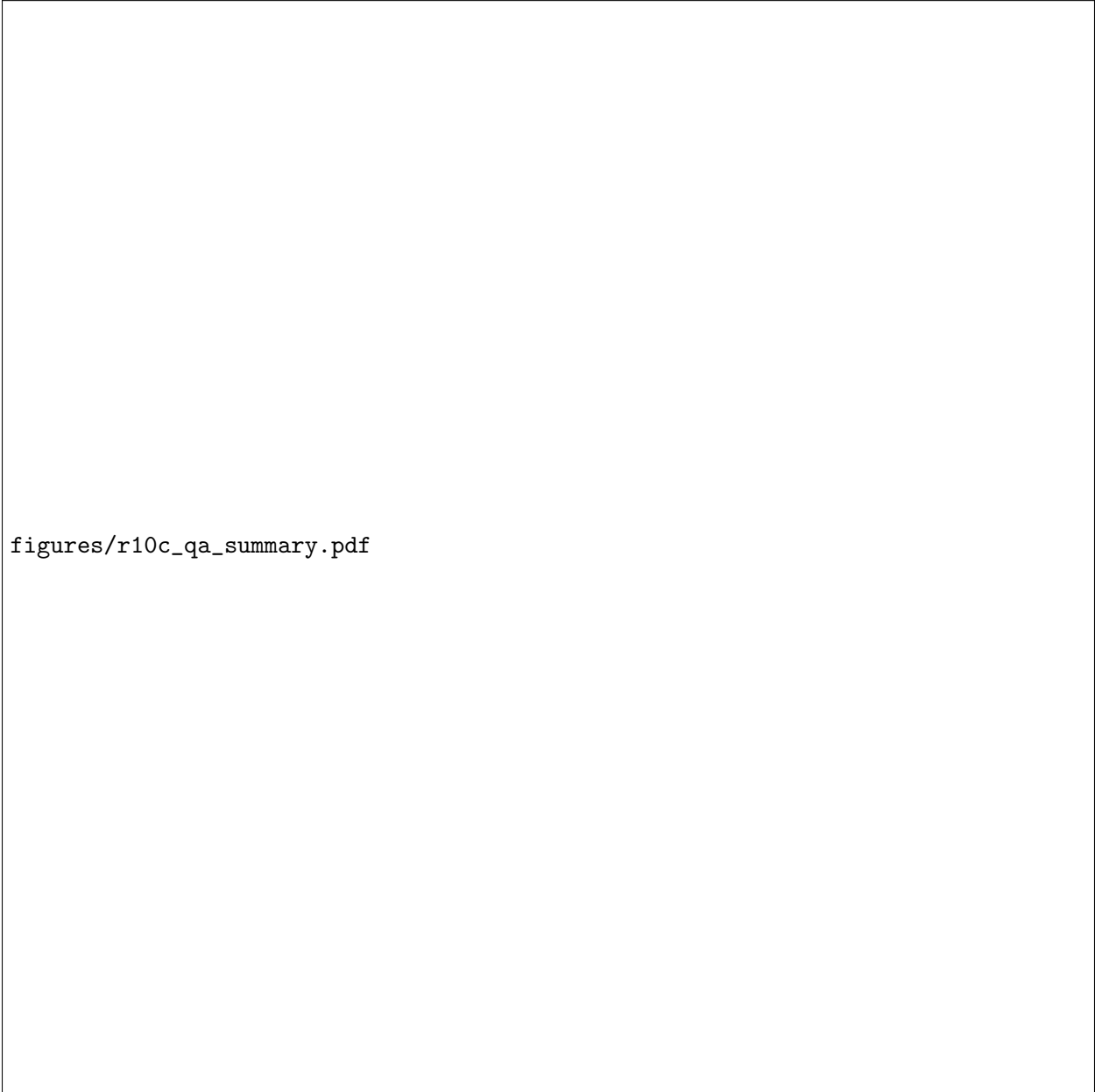
SpectralBridge is available from the Earth Lab repository and documentation site [5, 6]. This draft describes commit `c46f85a2af97974125161a23f5eb10f3b2c68a34`, dated 25 August 2026. Its metadata declares version 2.2.0, Python 3.10 or newer, and the GPL-3.0-or-later license. At that commit, continuous integration runs Python 3.11 on Ubuntu; other environments are not continuously tested [16].

Version 2.2.0 is development metadata rather than a matching archived release at the inspection date. DOI 10.5281/zenodo.11167877 identifies a 9 May 2024 pre-rename **cross-sensor-cal** archive, not the inspected SpectralBridge tree. A submitted paper should cite a permanent post-rename archive after version, authorship, and release metadata are reconciled [26, 27].

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`figures/r10c_qa_summary.pdf`

Figure 3: R10C shows the meaning of a completed workflow with an overall WARN. Passing diagnostics, warnings, and unevaluated scientific questions coexist in the same record [23]. Generated from a structured extract by `scripts/make_r10c_qa_figure.py`.